

Supplementary File

This **Table 7** summarizes the performance of eight machine learning models evaluated under three feature scenarios: (1) using only CDR and MMSE, (2) excluding CDR, and (3) excluding MMSE. Performance metrics include Accuracy, Kappa, Sensitivity, Precision, F1 Score, and AUC for both cross-validation (CV) and independent test sets. Results show that models using both CDR and MMSE achieve near-perfect performance. Excluding CDR leads to a substantial decline in performance, highlighting its importance as a key predictor. In contrast, excluding MMSE has a relatively minor effect, suggesting that CDR contributes more strongly to the model’s predictive power.

Table 7: Performance of ML models evaluated under three feature scenarios.

Scenario	Model	CV Acc	CV Kappa	CV Sens	CV Prec	CV F1	CV AUC	Test Acc	Test Kappa	Test Sens	Test Prec	Test F1	Test AUC
CDR+MMSE	RF	0.99	0.98	0.98	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00
CDR+MMSE	SVM	0.99	0.98	0.98	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Exclude CDR	RF	0.74	0.48	0.71	0.77	0.74	0.82	0.93	0.87	1.00	0.88	0.94	0.97
Exclude CDR	SVM	0.81	0.62	0.69	0.92	0.79	0.86	0.73	0.47	0.67	0.77	0.71	0.83
Exclude MMSE	RF	0.99	0.98	0.98	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Exclude MMSE	SVM	0.98	0.97	0.98	0.98	0.98	0.99	1.00	1.00	1.00	1.00	1.00	1.00